

Omar Marghadi

AI & Industrial Engineering Graduate

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Location: France, Mobile Europe



EDUCATION

École Centrale de Lyon

M.Sc. Data & AI in Industrial Engineering (DIAGI)

Grade: Merit (Mention Bien) / Thesis: Magna Cum Laude (15/20)

Lyon, France

2024 – 2025

Arts et Métiers (ENSAM)

Engineering Degree – Data Science and AI Track

Selected for Centrale Lyon's Double Degree Program

Morocco

2020 – 2025

International Baccalaureate

Mathematics & Engineering Science

GPA: 4.0 (Highest Honors: 17.65/20)

Morocco

2020

PROFESSIONAL EXPERIENCE

Orange Innovation labs – R&D Engineer Intern (Master's Thesis)

Lannion, France | Apr – Oct 2025

- Architected ROS2 perception pipeline integrating Valetudo API with RoboRock robots for multi-floor localization
- Developed multi-floor map alignment algorithm achieving 73.8% IoU using morphological preprocessing
- Implemented Graph Attention Networks (GAT) for spatiotemporal anomaly detection, improving F1-Score by +11%
- Fused multimodal IoT sensor streams (LiDAR, WiFi RSSI, environmental sensors) for context-aware alert generation
- Integrated AI perception modules into Vue.js 3 interface for seamless deployment

Capgemini Engineering – R&D Engineer Intern

Casablanca, Morocco | Jun – Aug 2024

- Architected LangGraph multi-agent framework orchestrating specialized tools for automated decision-making
- Built custom ETL pipeline with multi-format parsers (PDF, DOCX, XLSX, PPTX) and ChromaDB vector database
- Delivered complete POC with Chrome/Edge extension, demonstrating 60% efficiency improvement
- Technologies: LangGraph, LangChain, RAG, Llama 3.1 405B, Mistral Nemo 12B, NVIDIA NIM API

STELLANTIS & Centrale Lyon – Research Intern

Lyon, France | Jan – Mar 2025

- Conducted bibliometric analysis of ML applications to NVH numerical models for electric vehicles
- Analyzed patents and data-driven approaches to replace costly empirical simulations

3D Smart Factory – Deep Learning Research Intern

Remote | Jun – Sept 2024

- Explored and adapted Kolmogorov Arnold Networks (KAN) for 3D CT-scan image segmentation
- Conducted comparative analysis against baseline CNN architectures for medical imaging applications

OCP Group – Industrial Engineering & Computer Vision Intern

Khouribga, Morocco | Jun – Jul 2023

- Implemented real-time human pose estimation (HRNet) for ergonomics and safety monitoring
- Designed supervision architecture for CNC equipment monitoring and maintenance digitalization
- Analyzed GMAO automated processes and identified optimization opportunities

PROJECTS & HACKATHONS

Drone Defense Hackathon – "D.U.E.L" Multi-Drone System – Finalist, ADOPT AI Summit 2025

Autonomous Quadcopter Platform for B2B Solutions (STM32, Jetson Nano, Isaac Sim – Ornis Startup (Ongoing)

AI for Robotic Control – Master's Research Project (Bibliography + Bibliometry)

Bio-Inspired Seal Robot for Ocean Sensing (4-DOF locomotion and ESP32 control) – Centrale Lyon, Eco-Design Project

Desktop Buddy: VLA Self-Balancing Robot – Personal Project (In Progress)

LLM-Enhanced FMEA Analysis – Lund University Collaboration

Autonomous Vehicle for Inventory Management – 2nd Prize, African Robotic Brain

Campus AR Navigation System – Cloud-Native Application

Industry 5.0 Smart Factory & Digital Twin – Arts et Métiers Project

TECHNICAL SKILLS

Robotics & Control	ROS 2, MuJoCo, Gazebo, AirSim, Unreal Engine, MoveIt2, sensor fusion, trajectory planning, SLAM
AI & Machine Learning	PyTorch, TensorFlow, Deep RL (DQN, PPO, SAC), GAT, KAN, Computer Vision, Imitation Learning, Transformers
Computer Vision	YOLO, OpenCV, Visual SLAM, HRNet, MediaPipe, 3D segmentation, object detection
Embedded Systems	Arduino, ESP32, ESP8266, Raspberry Pi, STM32, NVIDIA Jetson Nano, embedded C/C++
Programming	Python, C/C++, JavaScript, MATLAB, Bash
Frontend Development	Vue.js 3, React, modern JavaScript, responsive design, dashboard development
Backend Development	Flask, FastAPI, REST APIs, microservices architecture, Node.js
Data Engineering	ETL pipelines, SQL, NoSQL (MongoDB), ChromaDB, vector databases, data processing
DevOps & Cloud	Git, Docker, Kubernetes, CI/CD, Linux, GCP, Azure, Firebase
LLMs & GenAI	LangChain, LangGraph, RAG, fine-tuning, prompt engineering, multimodal AI
Industrial	SCADA, CNC automation, Industry 4.0, IoT, FMEA, Lean, Six Sigma

LANGUAGES & CERTIFICATIONS

Languages: English (C1 – IELTS 7.0), French (Bilingual), Arabic (Native)

Certifications & Awards:

- Machine Learning Specialization (DeepLearning.AI / Stanford)
- Six Sigma Yellow Belt Professional
- Finalist – Drone Defense Hackathon 2025 (ADOPT AI Summit, Grand Palais Paris)
- 2nd Prize – African Robotic Brain 2024
- Top 8/200 – Think AI Morocco Hackathon 2024
- LeRobot HuggingFace Hackathon 2025 Participant

Leadership & Activities:

- Organizer – Corporate Forum (21st edition, 32 companies)
- President – WEDESIGN Workshop (Digital Art & Mentoring)
- Active Member – AI & IT student clubs
- Competitive Programming – AMPC Competition

Interests: Tech News & Conferences, 3D Design & Modeling, Open-source robotics, Football & Swimming